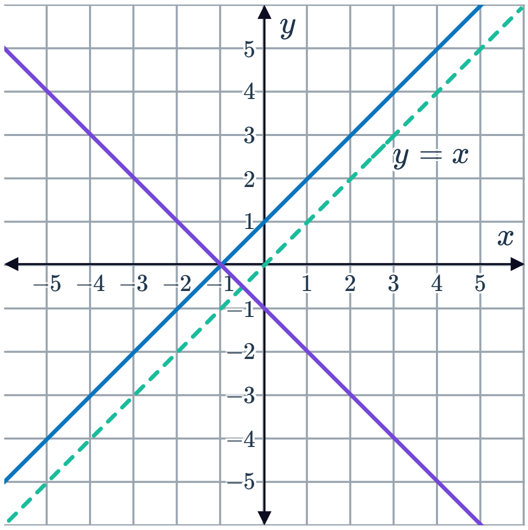
Inverse functions

1 Explain how a graphing utility such as a graphing calculator be used to visually determine if two functions are inverses of each other.

2 In each graph below two functions and the line $y = x$ are drawn. For each graph, state whether the functions are inverse functions of each other.

a



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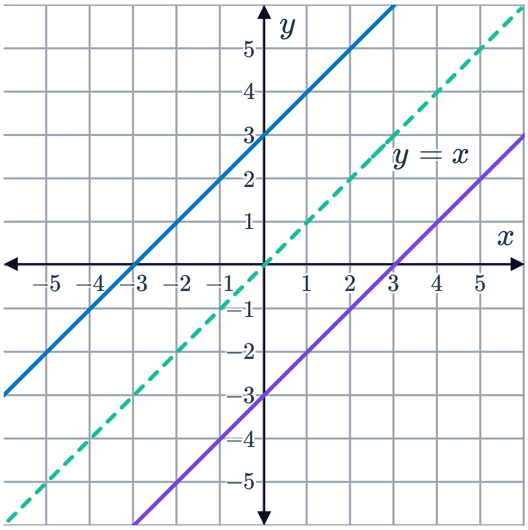

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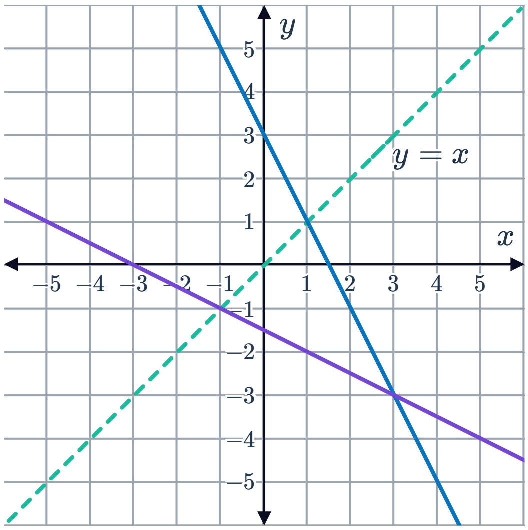

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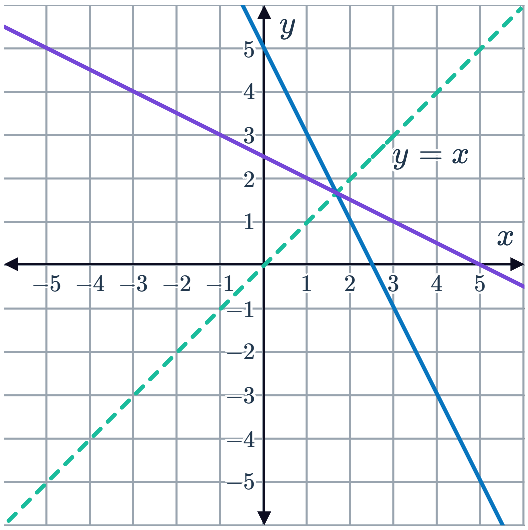

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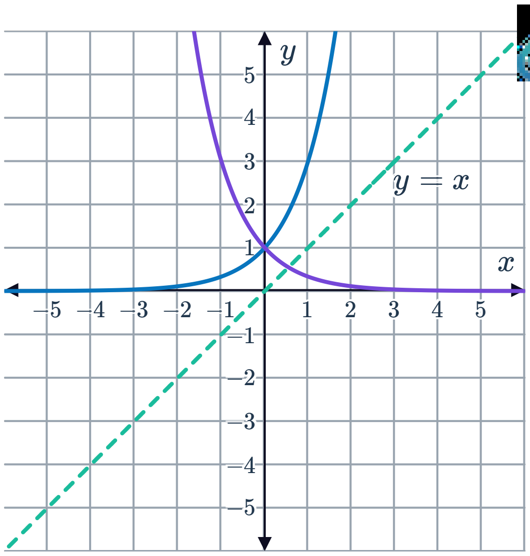

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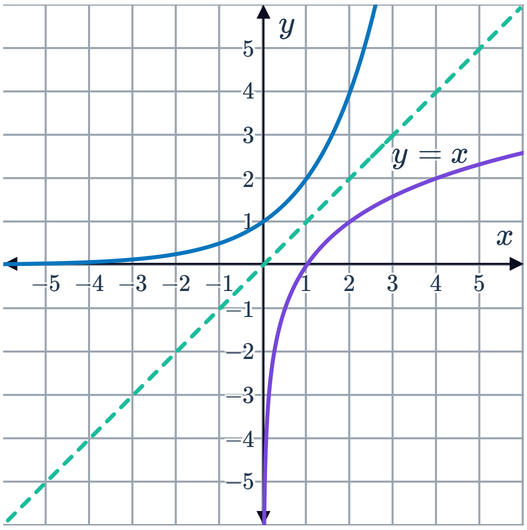

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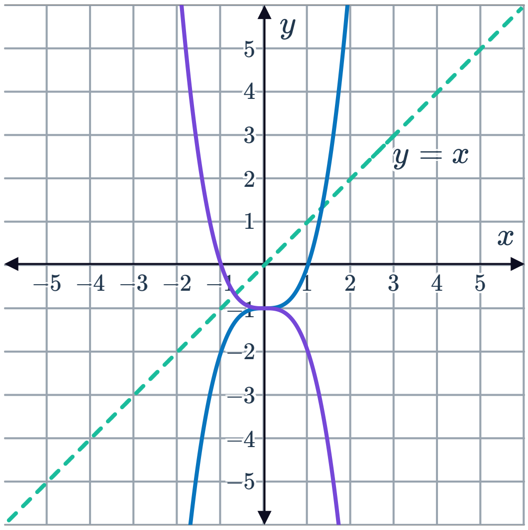


3 For each of the following functions, find an expression for the inverse function $y = f^{-1}(x)$:

a $f(x) = 7x - 8$

b $f(x) = \dfrac{4x}{3} - 5$

c $f(x) = 6x^3$

d $f(x) = \sqrt[3]{x-2}$

e $f(x) = \dfrac{8}{x-9} + 5$

f $f(x) = (x-4)^3 + 3$

g $f(x) = \dfrac{8}{5x+3}$

h $f(x) = \dfrac{7}{(x+8)^3} - 3$

i $f(x) = 4(9x-2)^3 - 7$

4 For each of the following functions, find:

i $f^{-1}(x)$

ii The domain of $f^{-1}(x)$

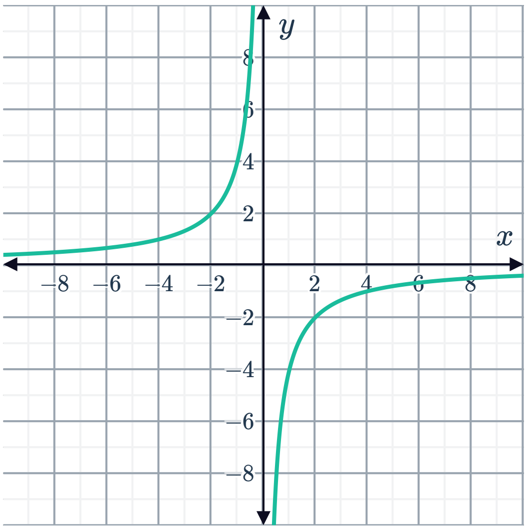
iii The range of $f^{-1}(x)$

a $f(x) = 8x - 9$ defined over $[-4, 2]$

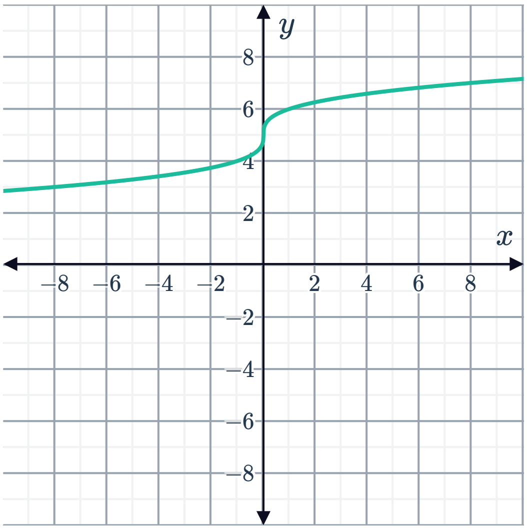
b $f(x) = x^2$ defined over $[0, \infty)$

5 For each of the following functions, find an expression for the inverse function $y=f^{-1}(x)$:

a $f(x)=-\frac{4}{x}$



b $f(x)=\sqrt[3]{x}+5$



Graphs of inverse functions

6 Consider the functions $f(x)=\frac{1}{x}+3$ and $g(x)=\frac{1}{x-3}$.

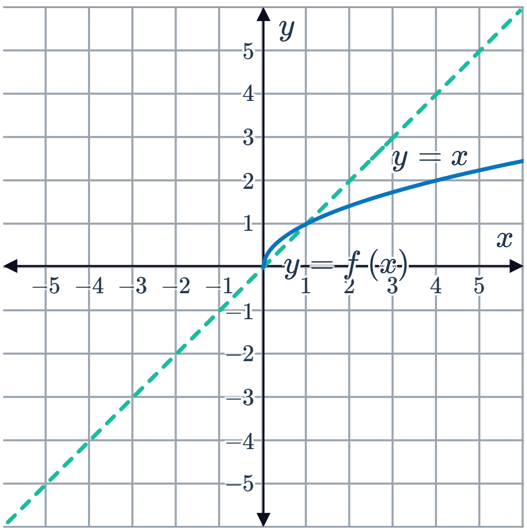
a Sketch the graph of $f(x)$.

b Sketch the graph of $g(x)$ on the same set of axes.

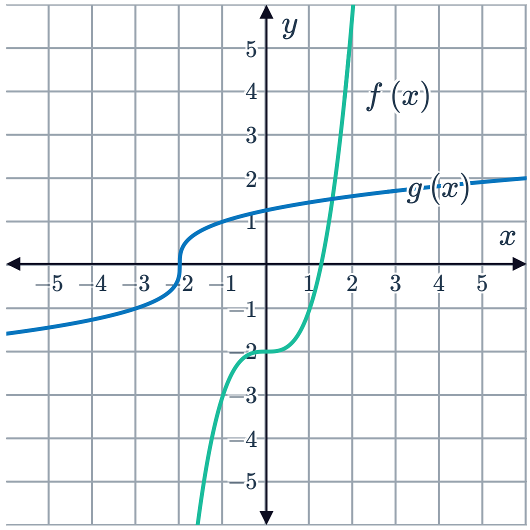
c Are $f(x)$ and $g(x)$ inverses?

7 Consider the graph of the function $f(x)$ over the line $y=x$:

Sketch the graph of $f^{-1}(x)$.



8 Consider the graphs of $f(x)$ and $g(x)$:



a State the equation of $f(x)$.

b State the equation of $g(x)$.

c Evaluate $f(g(x))$.

d Evaluate $g(f(x))$.

e State whether the following statements are correct:

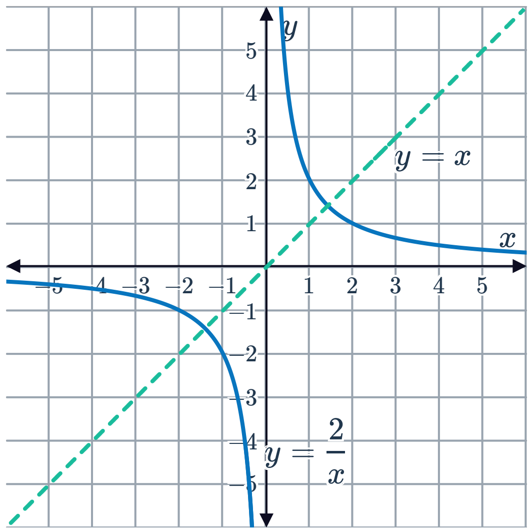
i $g(x)$ is an inverse of $f(x)$.

ii $f(g(x))$ has gradient -2.

iii $f(x)$ is an inverse of $g(x)$.

iv $g(f(x))$ has gradient 1.

9 Consider the graph of $y = \frac{2}{x}$ over the line $y = x$:

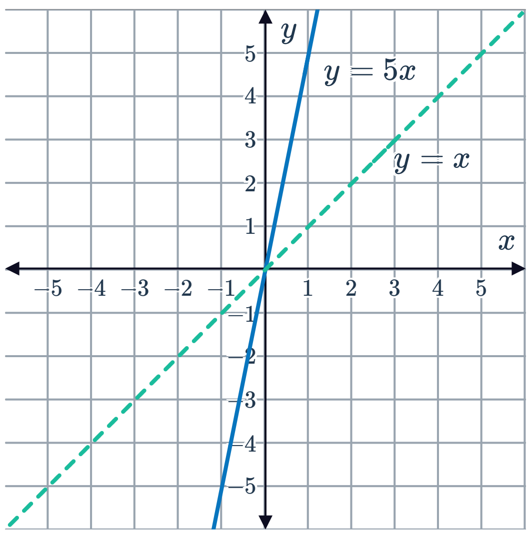


a Sketch the graph of the inverse of $y = \frac{2}{x}$.

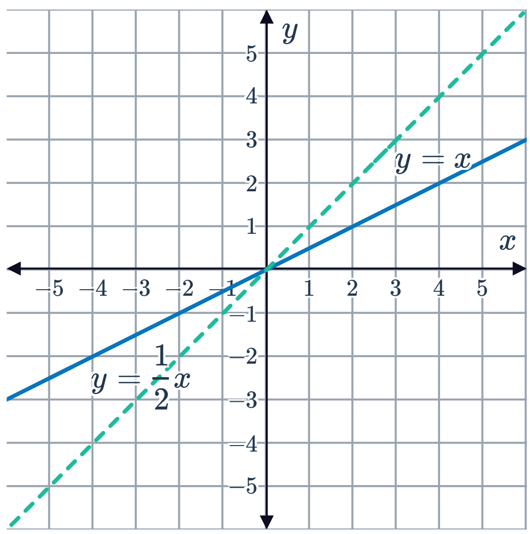
b Compare the inverse graph to the original graph.

10 Sketch the graph of the inverse of the following functions:

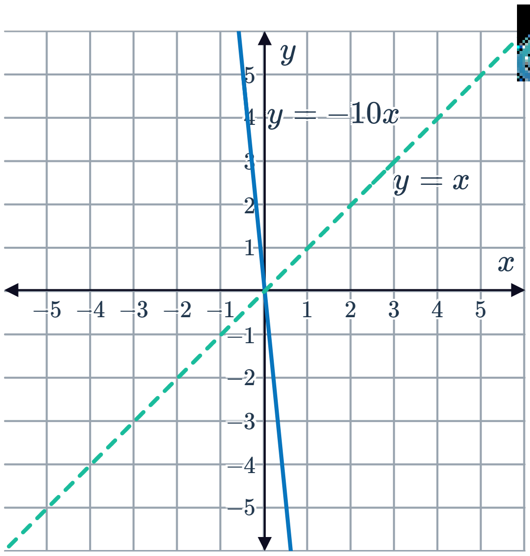
a



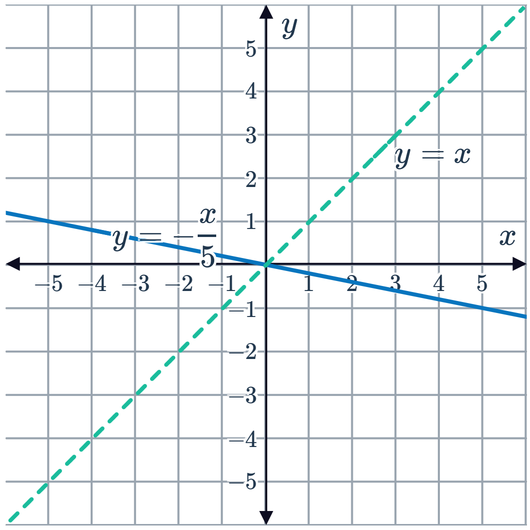
b



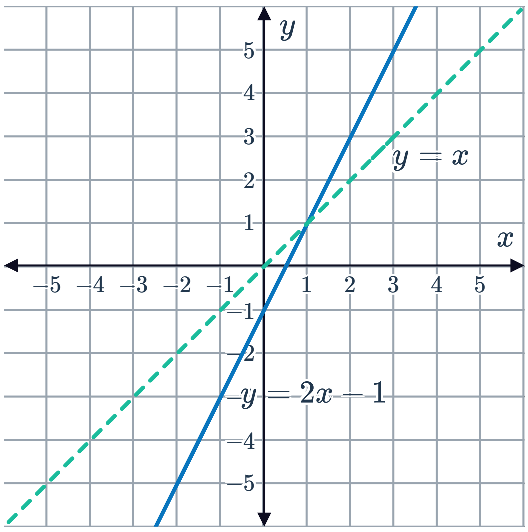
c


$y = -10x$
$y = x$
x
y

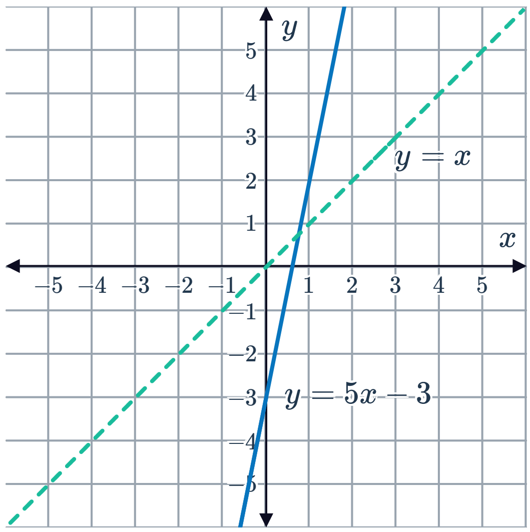
d


$y = x$
$y = -\frac{x}{5}$
x
y

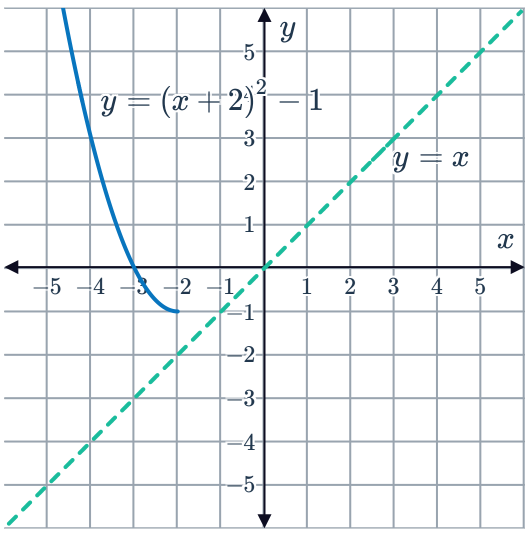
e


$y = x$
$y = 2x - 1$
x
y

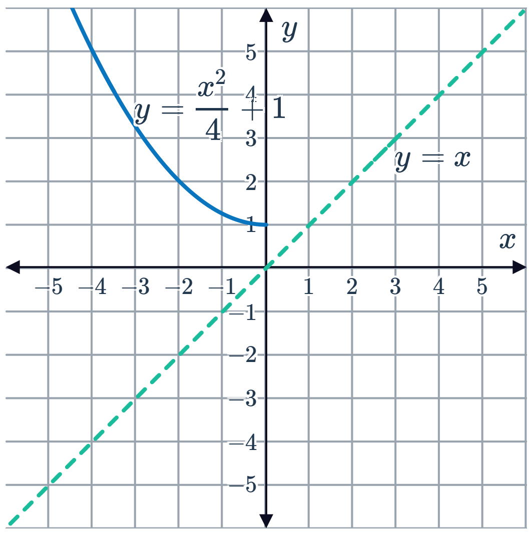
f


$y = x$
$y = 5x - 3$
x
y

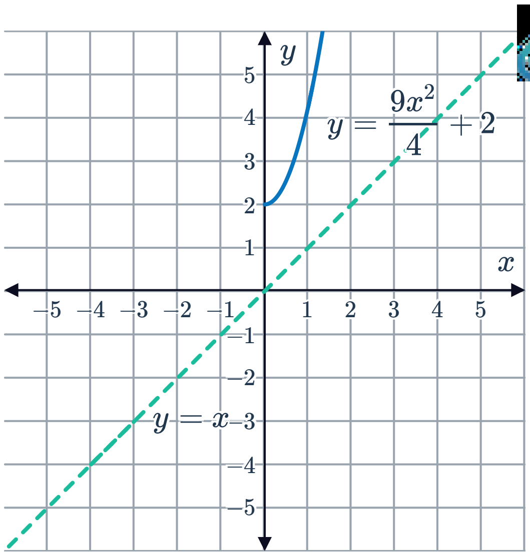
g


$y = (x + 2)^2 - 1$
$y = x$
x
y

h


$y = \frac{x^2}{4} + 1$
$y = x$
x
y

i



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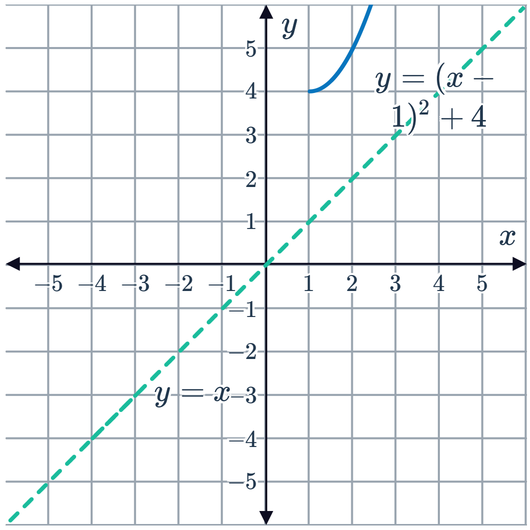

11 For each of the following functions:

i Sketch the function $f(x)$ over its domain.

ii Find the inverse, f^{-1}.

iii State the domain of f^{-1}.

iii State the range of f^{-1}.

iv Sketch the function f^{-1} over its domain.

a $f(x) = x + 6$ defined over the interval $[0, \infty)$.

b $f(x) = 7 - x$ defined over the interval $[2, 9]$.

c $f(x) = (x - 6)^2 - 2$ defined over the interval $[6, \infty)$.

d $f(x) = \sqrt{4 - x}$ defined over the interval $[0, 4)$.

e $f(x) = (x + 2)^2 + 3$ defined over the interval $[0, \infty)$.

12 Consider the functions $f(x) = x^2 - 5$ and $g(x) = \sqrt{x + 5}$, for $x \geq 0$. The function y is defined as $y = g(f(x))$, for $x \geq 0$.

a State the equation for y.

b Graph the functions $f(x)$, $g(x)$ and y on the same set of axes.

c What do you notice about the graph of y in relation to the graphs of $f(x)$ and $g(x)$?

Applications

13 The function $t=\sqrt{\frac{d}{4.9}}$ can be used to find the number of seconds it takes for an object in Earth's atmosphere to fall d metres.

- **a** State the function for d in terms of t.
- **b** Find the distance a skydiver has fallen 5 seconds after jumping out of a plane.

14 The following formula can be used to convert Fahrenheit temperatures x to Celsius temperatures $T\left(x\right)$:

$$T\left(x\right)=\frac{5}{9}\left(x-32\right)$$

- **a** Find $T\left(-13\right)$.
- **b** Find $T\left(86\right)$.
- **c** Find $T^{-1}\left(x\right)$.
- **d** What can the formula T^{-1} be used for?

15 A particle is moving along a straight line path. After t seconds, its velocity is given by the equation $v=\left(96t-80\right)^2$.

- **a** Solve for the time t at which the particle comes to rest.
- **b** Determine the equation for time, t, in terms of velocity, v, that represents the motion of the particle before it has come to rest.

16 The tax on a new tablet is 7% of the advertised price, A.

- **a** Determine the equation for the total cost T as a function of the advertised price A.
- **b** Hence, express the advertised price A as a function of the total cost T.

17 The function $T=2\pi\sqrt{\frac{l}{9.8}}$ can be used to find the period T of a simple pendulum of length l metres.

- **a** Express the length l as a function of the period T.
- **b** Hence, find the length of a pendulum whose period is 1.5 seconds. Round your answer to one decimal place.